Statistical Analysis Plan (SAP v2.1)

Pre-specified Statistical Pipeline & Analytical Protocol (Post-Review R1 Revision)

Study Title	Exploratory Survey of Herbal Medicine Use and Safety Awareness among Digitally Connected Syrian Adults
Target Journal	PLOS ONE (Manuscript Revision R1)
Document Version	v2.1.0 (Final Post-Peer Review Specification)
Ethical Approval	Biomedical Committee, SHAMNA (Ref: SHAMNA-26-04, 10 April 2026)
Primary Analytical Dataset	Anonymized_Dataset_N_287.csv (N = 287 eligible adults)

1. Administrative & Study Background

This Statistical Analysis Plan (SAP v2.1) details the formal quantitative methodologies, decision rules, statistical tests, and reproducible software environment applied to evaluate the primary survey dataset (N = 287). This revised plan strictly implements all methodological upgrades mandated during peer review at PLOS ONE, establishing 100% alignment between raw analytical scripts, tabulated outputs, and manuscript text.

2. Primary Research Hypotheses & Testable Objectives

- **Hypothesis 1 (Herb-Drug Interaction Awareness):** Healthcare background (Q5) is positively associated with self-reported awareness of herb-drug interaction risks (Q12). Tested via 2x2 contingency table using Yates' continuity correction, Fisher's exact test, and Odds Ratio (OR) calculation.
- **Hypothesis 2 ('Natural = Safe' Belief):** Educational attainment level (Q4) is associated with the perception that natural products are inherently completely safe without risk of harm (Q13). Tested via 4x5 contingency table using Pearson Chi-Square and Monte Carlo exact p-value simulation.
- **Hypothesis 3 (Unconsulted Herbal Usage Frequency):** Healthcare background (Q5) influences the frequency of self-administered herbal medicine use without professional medical or pharmacological consultation (Q22). Tested via 2x5 contingency table with Cramér's V effect size.

- **Hypothesis 4 (Digital Health App Adoption Interest):** Healthcare background (Q5) influences public willingness to adopt a free Arabic-language home assistant mobile application for verified medicinal herb information (Q29). Tested via 2x4 contingency table with Monte Carlo exact simulation.

3. Sample Specification & Data Cleaning Protocol

The raw data collection yielded N = 300 total survey entries. In accordance with ethical approval SHAMNA-26-04, a pre-planned exclusion rule was executed to remove underage respondents (< 18 years old, n = 13), establishing the finalized analytical dataset of N = 287 eligible adult participants. All closed-ended items featured mandatory response validation, eliminating missing data across primary survey variables.

4. Inferential Statistical Testing Pipeline

Hypothesis / Test	Variables	Table Dim.	Primary Test Method	Effect Size / Metrics
Test 1: Interaction Awareness	Q5 (Healthcare) x Q12 (Awareness)	2 x 2	Yates' Chi2 (X2=6.17, p=0.013) & Fisher's Exact (p=0.010)	OR = 2.14 (95% CI: 1.20-3.80), Cramér's V = 0.155
Test 2: Natural=Safe Belief	Q4 (Education) x Q13 (Belief)	4 x 5	Pearson Chi2 (X2=16.73, p=0.160) & Monte Carlo p=0.158	Cramér's V = 0.139 (Sparse expected cells < 5)
Test 3: Unconsulted Use	Q5 (Healthcare) x Q22 (Frequency)	2 x 5	Pearson Chi2 (X2=4.07, p=0.396) & Monte Carlo p=0.392	Cramér's V = 0.119
Test 4: App Willingness	Q5 (Healthcare) x Q29 (Willingness)	2 x 4	Pearson Chi2 (X2=3.32, p=0.344) & Monte Carlo	Cramér's V = 0.108

p=0.355

5. Dual Sensitivity Analysis Protocol

To verify robust data integrity against sampling anomalies, two pre-planned sensitivity analyses were conducted (detailed in S2 Appendix):

- **Sensitivity Analysis 1 (N = 286):** Exclusion of 1 suspect rapid duplicate entry (8-second timestamp gap, rows 225-226). Results confirmed maximum descriptive variation < 0.4% and 100% statistical decision concordance.
- **Sensitivity Analysis 2 (N = 258):** Exclusion of all pilot survey participants (n = 29 eligible pilot respondents). Results confirmed maximum descriptive shift < 1.9% and identical p-value decision thresholds across all inferential tests.

6. Reproducible Execution Environment

All analytical computations were migrated from proprietary tools to an open-source Python 3.11+ environment. The complete analysis pipeline is executable via **reproduce_analysis.py** using standard libraries (pandas v2.0+, NumPy v1.24+, SciPy v1.10+). Monte Carlo exact p-values were generated using 10,000 resamplings with a fixed random seed (**20260415**) to guarantee bit-for-bit numerical reproducibility.